

A Memory That Only Looks Like One: The Recurrent State of a Hybrid Vision-Language Model Relays the Picture Rather Than Storing It

Vizuara Research

August 2026

Abstract

In a transformer an image persists: it becomes hundreds of key-value entries that any later computation can attend to. In a hybrid linear-attention model it does not. Most layers compress what they read into a fixed-size recurrent state that later tokens overwrite. It is already known that recurrent models are worse than transformers at retrieving from context, and that a fixed-size state is the reason. What is not known is *when* a particular model forgets, *what* it keeps while forgetting, and *what breaks* downstream as a consequence. We answer all three for Zamba2-VL-7B, whose 81 decoder layers all carry a Mamba2 mixer while only 13 additionally carry attention.

First, at the moment of writing the recurrent state is highly informative about *which* object is present and close to uninformative about *how many* or *where*: it records gist, not particulars. A probe on the best single layer reads object identity at 92.5% against a 12.5% chance rate.

Second, the horizon over which that state genuinely retains its own past can be computed before any experiment. Mamba2 multiplies the state by $\exp(\Delta_t A)$ at every step, so retention after k tokens is $\exp(A \sum \Delta_t)$ and the per-head half-life is $\ln 2 / (|A| \bar{\Delta})$. Across three model scales the median predicted half-life is 6.6, 2.9 and 2.6 tokens, with nothing fitted.

Third, and this is the paper’s central result, what a probe reads from that state at distance is largely not retention. Splitting heads into thirds by predicted half-life inverts the expected ordering: heads with a two-token half-life decode identity at 61.7% after sixty-four tokens, which retention cannot produce. Evicting the visual key-value entries immediately after the image, before any subsequent token is processed and without touching the recurrent state, drops those heads from 51.6% to 7.8% at thirty-two tokens, below chance, while leaving slow heads at distance zero nearly unaffected. How much a head’s readout depends on the attention channel is itself predicted by that head’s decay rate, correlating at $r = 0.991$ across the three strata. The recurrent channel relays what attention is holding rather than storing it, and the weights say which heads relay most.

This resolves the causal result. Exchanging all 81 recurrent states between two runs that saw different images leaves 100 of 100 answers following the host’s own image, with every swap verified, whereas exchanging seven of the thirteen attention caches moves 94 of 100 to the donor’s image. The recurrent swap cannot matter because it is overwritten within a few tokens from the channel it does not touch. Finally, the redundancy premise behind visual token and key-value pruning survives on this architecture down to a quarter of the entries, but the cost of pruning harder roughly doubles with distance: removing every visual entry costs 34 accuracy points beside the image and 59 at thirty-two tokens.

1 What Was Known Before, and What Is New Here

Linear-attention and state-space sequence models replace the growing key-value cache of a transformer with a state of fixed size [1, 2, 3, 4, 5, 6]. Because a fixed state cannot grow with the context, most

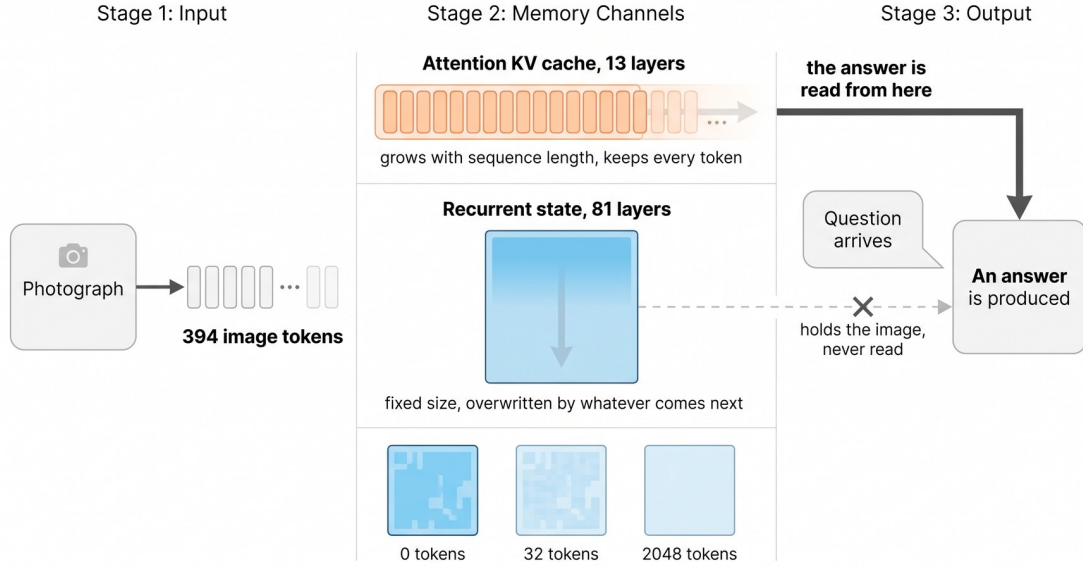


Figure 1: The two memory channels of a hybrid vision-language model and the question this paper asks of them.

deployed systems keep a few attention layers alongside it, and the resulting hybrids are now common [7, 8, 9, 10, 11]. This paper asks what the recurrent half of such a model is actually holding.

The premise that a fixed-size recurrent state must eventually lose information from context is not ours and we do not claim it. Jelassi et al. [12] prove that transformers can copy exponentially long strings while generalized state space models are, in their words, “fundamentally limited by their fixed-size latent state”, and show empirically that pretrained transformers “dramatically outperform state space models at copying and retrieving information from context”. Related results sharpen the same point from other directions: state-space models are limited to TC^0 and cannot reliably track state [13], in-context retrieval is the specific bottleneck separating recurrent models from transformers [14], and most of the quality gap between efficient architectures and attention is attributable to recall [15, 16]. Any reader who knows these results should read our work as building on them rather than rediscovering them.

Table 1 states what that literature establishes and what this paper adds.

2 Significance

Hybrid linear-attention models are being adopted because a fixed-size state promises constant-memory inference. The design question that follows is which work the recurrent channel can actually be trusted with. This paper turns that from a matter of judgement into a calculation: given a checkpoint’s A and its gate statistics, the number of tokens over which its recurrent state retains what it wrote can be computed before any experiment is run. For Zamba2-VL-7B that number is single digits, which is far shorter than the intuition invited by a state of 458,752 numbers per layer.

The consequence for practice is concrete. A substantial line of work accelerates vision-language models by discarding visual tokens or evicting their key-value entries, on the stated premise that those entries are largely redundant [24, 25, 26], alongside a parallel line that evicts key-value entries in language models on similar grounds [27, 28, 29]. Those methods were developed and validated on dense transformers, where they work, and we find they largely survive the move to this hybrid: a quarter of the

Established by prior work	New here
A fixed-size state must lose information from context, with a capacity proof and behavioural evidence [12, 13, 14].	<i>When.</i> A retention horizon in tokens, computed from the model’s own A and Δ with nothing fitted, at three scales.
Recurrent models underperform at copying and retrieval as a class [15, 16].	<i>What survives.</i> The failure is content-selective: object identity is highly decodable at the same instant that count and position are near chance.
Architectures are compared to one another from the outside.	<i>Causal localisation inside one model,</i> and the finding that the recurrent channel relays rather than stores, which makes probing it at distance a measurement of the attention channel.
Visual tokens and key-value entries carry substantial redundancy that can be pruned [24, 25, 26].	<i>What underwrites that premise,</i> and that its safety margin shrinks with distance from the image on a hybrid.

Table 1: Delineation of prior work and contribution.

visual entries is enough to hold accuracy at both distances we tested. What does not survive is the safety margin. The recurrent channel underwrites that redundancy by relaying the picture, and it only relays for a few tokens, so the cost of aggressive pruning roughly doubles between a prompt that asks immediately and one that asks thirty-two tokens later. A method tuned and validated on short prompts will therefore look safer than it is, which is the failure mode hardest to catch in benchmarking.

3 Model and Instruments

3.1 The model

Zamba2-VL-7B [35] is the vision-language member of the Zamba family [8], which pairs a Mamba2 backbone [5] with a shared attention module, and it takes its vision encoder from Qwen2.5-VL [31]. It is Apache-2.0 licensed and has 81 decoder layers. Every one carries a Mamba2 mixer, and 13 of them additionally carry a shared attention block, at layer indices 6, 11, 17, 23, 29, 35, 41, 47, 53, 59, 65, 71 and 77. The recurrent channel therefore spans all 81 layers and the attention channel spans 13. Each layer’s recurrent state is a tensor of shape $[112, 64, 64]$, which is 458,752 numbers. One image occupies between 345 and 415 tokens depending on its aspect ratio.

3.2 Reading the state

The Mamba2 recurrence is

$$S_t = S_{t-1} \exp(\Delta_t A) + \Delta_t x_t B_t^\top, \quad y_t = S_t C_t, \quad (1)$$

so writing uses (x, B) , reading uses C , and retention of anything already present is governed by A and Δ alone. We verified each of these against the installed model source rather than assuming the canonical form. In particular $A = -\exp(A_{\log})$ is a per-head scalar, the gate is $\Delta = \text{softplus}(\Delta_{\text{raw}} + b)$ clamped below at 0.001 with no upper clamp, and the decay factor is $\exp(\Delta_t A)$ elementwise.

Reading a representation with a trained classifier is standard [17], and so are its known hazards: a probe can succeed because it is powerful rather than because the information is used, which is why selectivity against a control task matters [18] and why probing results are read with care [19]. We

therefore pair every probe with an intervention, in the tradition of causal mediation analysis [20] and activation patching [21, 22, 23], and we follow the recommendation of that literature to report what the intervention does to behaviour and not only to the readout [23].

Because 458,752 numbers per layer is far too wide to probe directly, each layer’s state is passed through a fixed Gaussian random projection to 512 features. A Johnson-Lindenstrauss projection preserves linear separability up to a controlled distortion, so a linear probe on the projected features is, to that tolerance, a linear probe on the state. The projection columns are drawn independently, so the first k columns are themselves a valid k -feature projection.

3.3 Splicing the channels

The splice exchanges one memory channel between two runs that saw different images, leaving the other untouched, and then asks the host’s question. Swapping the recurrent channel replaces all 81 states and convolution buffers; swapping the attention channel replaces the 13 key and value caches. The host cache is restored exactly afterwards, so one prefill supports many conditions.

4 Materials

Trials are built from COCO val2017 instance annotations [34], the corpus underlying most visual instruction tuning [30], in three families: which object is present, how many of a named object are present, and where in the picture a named object is. All are eight-way forced choice, so the chance rate is exactly 12.5%. Trials are constructed in blocks of eight that share one identical candidate list, with each candidate correct exactly once inside the block. That construction is what makes the pre-image control interpretable: with the candidate set fixed and the answer balanced, a decoder that has not seen the image cannot exceed chance by exploiting a prior over candidates.

Images are fetched at native geometry for the decay experiments. For the splice they are letterboxed onto a common square canvas, because two runs on images of different shape emit different numbers of vision tokens and therefore different-length attention caches, which cannot be exchanged. Letterboxing rather than square resizing matters: in a pilot, a plain square resize deformed the content enough to halve identity accuracy, so every splice image is letterboxed.

5 Result 1: The State Records Gist, Not Particulars

At the moment of writing, the recurrent state is highly informative about which object is present and close to uninformative about how many or where. Probing the full state across all 81 layers reads identity at 83.1% against a 12.5% chance rate ($p = 6 \times 10^{-92}$, $n = 160$), count at 19.4% ($p = 0.009$) and position at 17.5% ($p = 0.041$).

Those figures understate what is linearly present. The full-state feature vector is $81 \times 512 = 41,472$ dimensions, which is heavily overparameterised relative to the number of trials, and a probe on a single layer does substantially better. Layer 64 alone reads identity at 92.5%, and the last 21 layers (60 to 80) average 85.2%, against 59.0% for the middle third of the stack (layers 27 to 53). On the identical trials the full-state probe reads 71.7%. The gist is therefore both stronger than the aggregate number suggests and localised in depth, concentrating in the late layers.

The recurrent state is not a compressed copy of the image. It is a selective summary that keeps object identity and discards count and position, and that selection is visible at the instant of writing rather than emerging over distance.

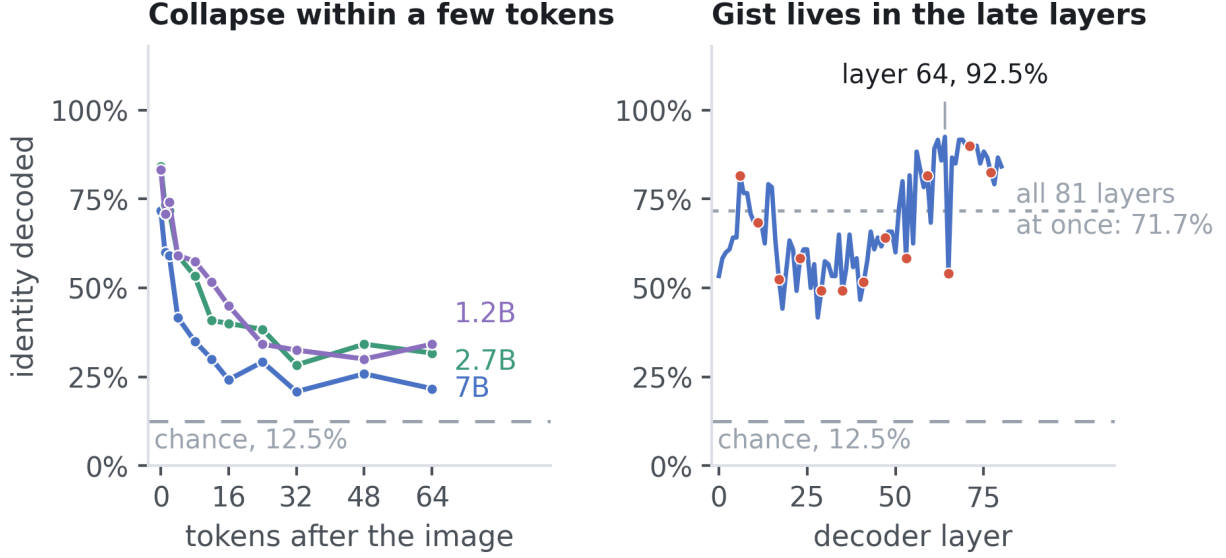


Figure 2: Left: identity decoded from the recurrent state against distance from the image, at three model scales, 120 trials per point. The pre-image control reads exactly 12.5% at every scale. **Right:** identity decoded from each layer’s state alone at the instant of writing, with red marking the 13 layers that also carry attention. A probe given all 81 layers at once does worse than the best single layer, because 41,472 features on 120 trials is heavily overparameterised.

6 Result 2: The Forgetting Horizon, Predicted From the Weights

6.1 The prediction

Retention of whatever the state held, after k further tokens, is

$$R(k) = \exp\left(A \sum_{t=1}^k \Delta_t\right), \quad \text{half-life} = \frac{\ln 2}{|A| \bar{\Delta}}. \quad (2)$$

A is a stored weight. $\bar{\Delta}$ is measured from the gate channels of the input projection over the tokens that follow the image, on the same prompts the decay experiment uses. Nothing here is fitted to a decay curve.

Across all $81 \times 112 = 9,072$ heads the predicted median half-life is 6.6 tokens, with an interquartile range of 2.8 to 17.8. 85.0% of heads fall below 32 tokens and 0.08% above 512. Measured over eight different photographs the median moves only between 6.44 and 6.86 tokens, so the horizon is a property of the model rather than of the stimulus.

6.2 The measurement

The original collection measured distances 0 and then 32, which is far outside the region where the prediction is specific. Table 2 shows a finer sweep on 120 identity trials.

The above-chance signal halves at 4.0 tokens, inside the predicted interquartile range of 2.8 to 17.8. Normalised to its value at distance zero, measured decodability correlates with predicted mean retention at $r = 0.973$ across the eleven distances, with no free parameters. Beyond 16 tokens the measured curve sits above the predicted median and between the median and the mean, which is what the heavy tail of

tokens	probe accuracy	p	behaviour	predicted retention (median)
0	71.7%	2×10^{-50}	94.2%	1.000
1	60.0%	2×10^{-34}	96.7%	0.901
2	59.2%	2×10^{-33}	97.5%	0.812
4	41.7%	1×10^{-15}	96.7%	0.659
8	35.0%	2×10^{-10}	97.5%	0.434
12	30.0%	3×10^{-7}	96.7%	0.286
16	24.2%	3×10^{-4}	97.5%	0.188
24	29.2%	1×10^{-6}	95.8%	0.082
32	20.8%	7×10^{-3}	95.0%	0.036
48	25.8%	6×10^{-5}	97.5%	0.007
64	21.7%	3×10^{-3}	98.3%	0.001

Table 2: Decodability of object identity against distance from the image, on 120 trials at every distance, with the retention predicted from the weights alongside. Chance is 12.5%. The pre-image control reads exactly 12.5%.

slow heads predicts: the residual signal at long distance is carried by the minority of heads with long half-lives.

Throughout this collapse the model’s own accuracy never moves. It answers correctly on 94 to 98% of trials at every distance, including those where the state has been emptied.

6.3 The within-model test, which inverts the prediction

A curve comparison cannot rule out coincidence, so we split the heads into thirds by predicted half-life and probed each third separately, using the same projection basis so the strata are comparable to one another and to the unstratified probe. The thirds have predicted median half-lives of 2.0, 6.6 and 29.0 tokens, a fourteen-fold span. Decay predicts that the slow third stays decodable furthest.

The opposite happened. Table 3 shows the fast third, whose predicted half-life is two tokens, decoding object identity at 61.7% after sixty-four tokens, while the slow third falls to 16.7% over the same span.

stratum	predicted $t_{1/2}$	$d=0$	$d=16$	$d=64$	signal kept at $d=64$
fast	2.0	87.5%	73.3%	61.7%	0.66
middle	6.6	81.7%	20.8%	15.0%	0.04
slow	29.0	68.3%	17.5%	16.7%	0.08

Table 3: Identity decodability by head stratum. “Signal kept” is the above-chance excess at $d=64$ as a fraction of the excess at $d=0$. The ordering is the reverse of the decay prediction.

The obvious confound runs the wrong way. Because the split is global rather than within-layer, the strata draw unequal numbers of heads from each layer, and depth matters a great deal for decoding. But the fast stratum is concentrated in *earlier* layers (mean layer 32.2, against 45.5 for the middle stratum), with only 10.1% of its mass in the strong late layers against 37.5% for the middle stratum, and a depth-weighted per-layer accuracy of 0.653 against 0.729. It sits in worse layers and still wins.

A head with a two-token half-life cannot be holding something written sixty-four tokens earlier: by then its state is composed almost entirely of recent writes. If such a head decodes object identity at 61.7%, the identity information in its state arrived recently, which means something else in the model is still supplying it.

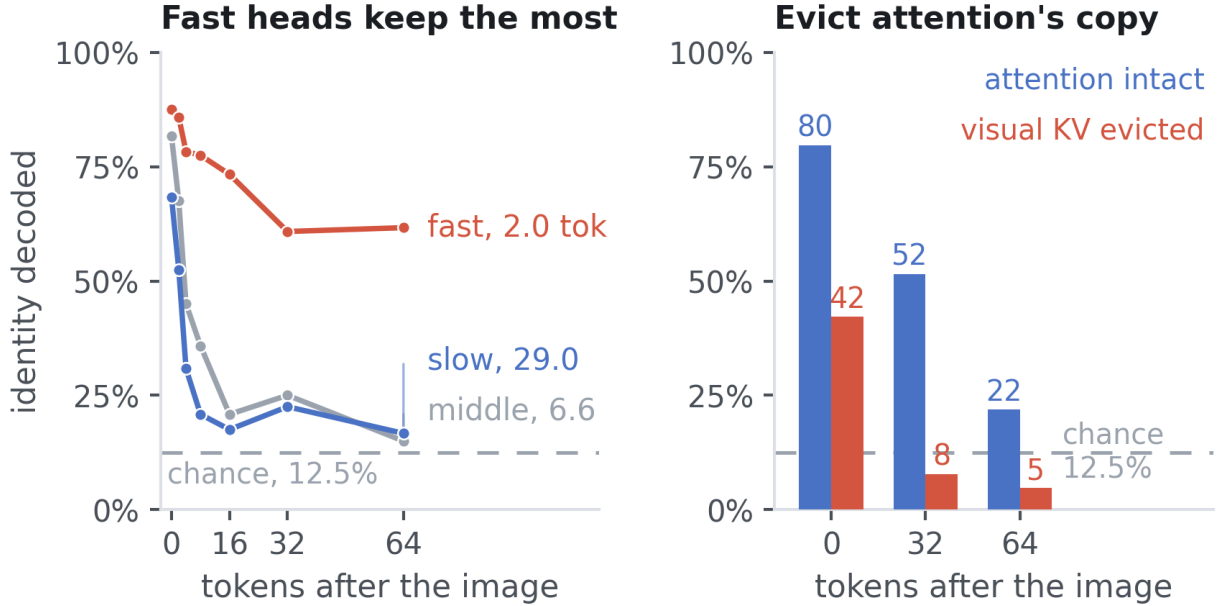


Figure 3: Left: heads split into thirds by the half-life implied by their own A and gate, labelled with each third’s predicted median. Decay predicts the slow third survives longest; the measured ordering is the reverse. **Right:** the same recurrent state probed with and without the attention channel’s copy of the image, fast heads only, $n = 64$ per bar. The eviction happens immediately after the image is read and never touches the recurrent state, which is identical in both runs at that instant.

7 Result 3: The Recurrent Channel Is a Relay, Not a Store

The only component that still holds the image at distance is the attention channel, whose key-value entries are untouched by distance. We tested that directly. Each trial was run twice, identically, except that in one run the visual key-value entries were evicted from all 13 attention layers immediately after the image was read and *before a single filler token was processed*. The eviction does not touch the recurrent state, which at that instant is identical in both runs.

If the state is remembering, eviction should not change what a probe reads from it. If the state is a relay, eviction should collapse decodability at distance, and should do so most severely for the fast heads.

stratum	distance	visual KV intact	visual KV evicted	change
fast	0	79.7%	42.2%	−37.5
fast	32	51.6%	7.8%	−43.7
fast	64	21.9%	4.7%	−17.2
middle	0	62.5%	43.8%	−18.8
slow	0	62.5%	54.7%	−7.8
all	0	59.4%	59.4%	0.0
all	32	6.2%	6.2%	0.0

Table 4: Identity decodability from the recurrent state with and without the attention channel’s copy of the image, $n = 64$ per cell. Chance is 12.5%. The fast heads’ apparent memory at 32 tokens is entirely dependent on the attention channel: removing it drops them from 51.6% to below chance.

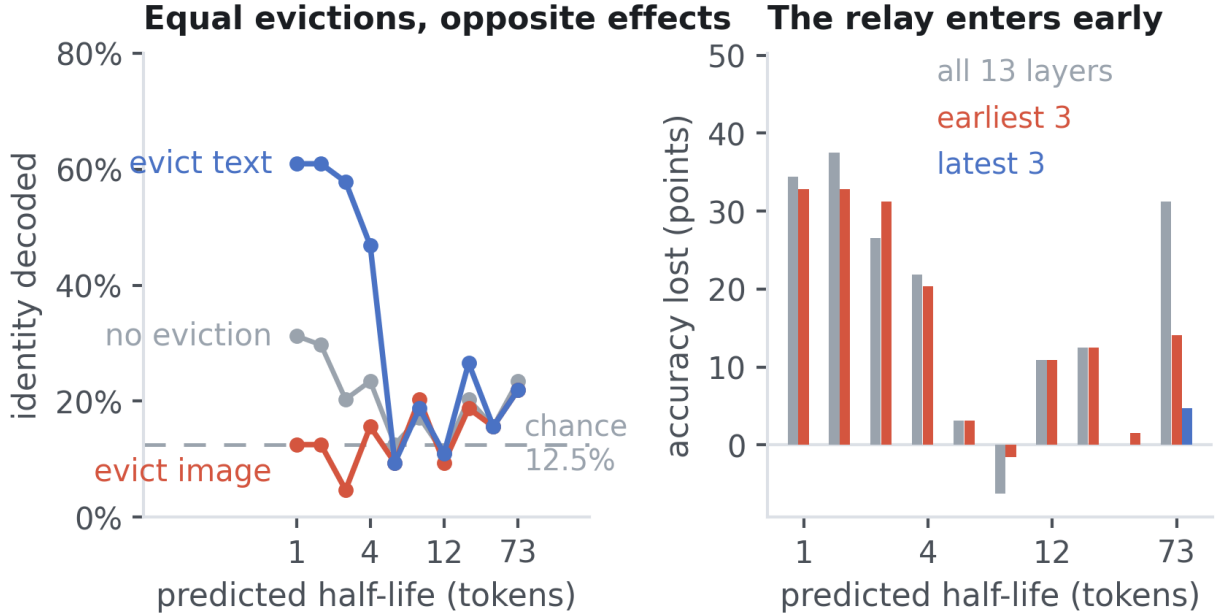


Figure 4: **Left:** the control that rules out the perturbation account. Both interventions remove the same number of positions from the same 13 caches, one taking the image and the other an equal count of the most recent pre-image text. They move the readout in opposite directions. **Right:** accuracy lost when the visual entries are zeroed at all 13 attention layers, at only the earliest three, and at only the latest three. The earliest three reproduce most of the full effect; the latest three do nothing.

7.1 Controls: the eviction removes the image, not merely positions

The standing objection to the eviction contrast is that it changes more than the image’s presence. Removing the visual entries shortens every attention cache by roughly 400 positions and changes the softmax normalisation for every token that follows. The matched control is to remove an equal number of *non-visual* positions from the same caches, and we ran it on ten strata formed by deciles of predicted half-life rather than thirds, with 64 trials per cell (Table 5).

The two interventions move the readout in opposite directions. At 32 tokens, evicting the image costs the fastest decile 18.8 points while evicting an equal number of the most recent pre-image text positions *raises* it by 29.7 points; averaged over the ten strata the two effects are -6.4 and $+12.5$ points. Removing text plausibly helps because it removes content competing with the image for the residual stream. A perturbation account predicts the same sign for both interventions, so this double dissociation rules it out.

The same collection locates where the relay enters. Zeroing the visual entries at only the three earliest attention layers reproduces 15.8 of the 17.2 points lost when all thirteen are zeroed, while the three latest layers produce 0.5 points. Visual information enters early and is consumed downstream. For per-layer conditions we zero rather than delete, because Zamba2’s attention blocks are shared and take a single mask length; the two operations agree to 2.34 points at $d=32$ where both are available, which is the validation of that substitution.

Finally, the contrast is stable across the choice of projection basis. Repeating it with three independent Gaussian projections gives, for the fast stratum at 32 tokens, intact $31.2 \pm 1.6\%$ against evicted $9.9 \pm 1.8\%$ (mean and standard deviation over seeds), while the slow stratum shows no eviction effect at any seed.

decile	pred. $t_{1/2}$	$d = 0$			$d = 32$	
		intact	evict image	evict text	intact	evict image
1	0.70	56.2%	18.8%	78.1%	31.2%	12.5%
2	1.84	56.2%	17.2%	79.7%	29.7%	12.5%
3	2.75	53.1%	18.8%	73.4%	20.3%	4.7%
4	3.79	46.9%	21.9%	71.9%	23.4%	15.6%
5	5.38	31.2%	26.6%	42.2%	12.5%	9.4%
6	8.15	35.9%	40.6%	51.6%	17.2%	20.3%
7	11.8	37.5%	25.0%	54.7%	10.9%	9.4%
8	17.8	45.3%	29.7%	57.8%	20.3%	18.8%
9	32.0	28.1%	26.6%	32.8%	15.6%	15.6%
10	73.3	51.6%	29.7%	46.9%	23.4%	21.9%

Table 5: The matched-eviction control by decile of predicted half-life, $n = 64$ per cell, chance 12.5%. “Evict image” and “evict text” delete the same number of positions from the same 13 attention caches. Single cells carry about 6 points of binomial noise at $n = 64$, and two of the twenty image-eviction cells fall slightly the wrong way; the trend rather than any one cell is the result. The drop under image eviction correlates with the decay rate $1/t_{1/2}$ at $r = 0.647$ at $d=0$ and $r = 0.800$ at $d=32$.

7.2 The decay parameters predict how much is relayed

Result 2 found that the parameter-derived horizon does not predict the measured decodability curve. It does predict something else, and the quantity it predicts is the one the relay account says it should. A head that forgets quickly must depend more heavily on fresh re-encoding, so its readout should be hurt more by removing the source of that re-encoding. Ordering the strata by decay rate gives exactly that:

stratum	predicted half-life	decay rate $1/t_{1/2}$	drop under eviction at $d=0$
fast	1.98	0.505	37.5 points
middle	6.64	0.151	18.8 points
slow	28.96	0.035	7.8 points

Table 6: Dependence on the attention channel, against the decay rate implied by each stratum’s own A and gate. The two quantities correlate at $r = 0.991$.

The decay calculation therefore does carry predictive content, but about the mechanism rather than about the curve: it says which heads are relaying and which are retaining. With three strata this is suggestive rather than established, and we flag it as such in the limitations.

The pattern is exactly the relay prediction and it is not subtle. At 32 tokens the fast heads read 51.6% with the attention channel intact and 7.8% without it, which is below the 12.5% chance rate. Thirty-three correct predictions out of sixty-four become five. Meanwhile the slow heads at distance zero, whose content was written while the image was being read, are barely affected by eviction at all (62.5% to 54.7%), which is the control that shows the intervention is not simply breaking the model.

What a probe reads out of the recurrent state at distance is not what the state remembered. It is what the attention channel is still supplying, re-encoded afresh at every token. The recurrent channel relays the picture; it does not store it.

7.3 What the probe can see, and why that matters here

The obvious objection to the eviction contrast is that the probe may simply be insensitive to weak signals. Freshly re-encoded content plausibly sits at higher amplitude than content written long ago and decayed, so a probe that finds strong signals and misses faint ones would produce these observations without the state having stopped storing anything.

That objection is quantitative and can be measured. Taking the distance-zero features, where the image is present at full strength, we attenuate each trial’s own contribution while substituting an equal amount of unrelated variation from another trial, holding total variance roughly constant: $x_i(\alpha) = \mu + \alpha(x_i - \mu) + \sqrt{1 - \alpha^2}(x_j - \mu)$. This is what decay followed by overwriting does to a stored trace.

The probe loses half its signal at $\alpha = 0.78$ and is indistinguishable from chance by $\alpha \approx 0.25$. The best single-layer probe behaves the same way, halving at $\alpha = 0.75$. So this instrument cannot see content below roughly a quarter of full amplitude.

That number does the work. The weights predict retention at thirty-two tokens of 3.7% at the median and 18% at the mean, both below the floor. Retained content at that distance therefore *could not* produce a visible reading. Yet the intact probe reads the fast heads at 51.6% there. Whatever it is reading cannot be a retained trace, because a retained trace at that distance is invisible to it. It must be high in amplitude, which means recently written. And removing the attention channel’s copy removes it.

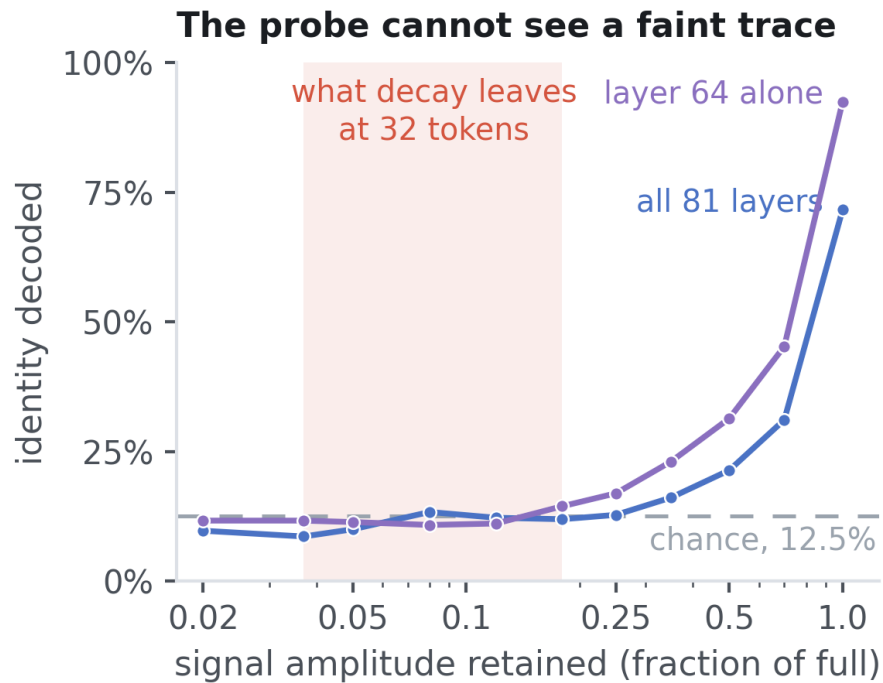


Figure 5: How faint a signal the probe can still recover. Each trial’s own contribution is attenuated while an equal amount of unrelated variation is substituted, holding total variance roughly constant. The shaded band is what the model’s decay parameters leave of a stored trace after 32 tokens, between the median and the mean over heads. Both probes are at chance throughout that band, so a visible reading at 32 tokens cannot be a retained one.

The probe’s insensitivity to faint signals is what makes the positive reading informative. A signal the instrument can see at thirty-two tokens is necessarily a fresh one, because a retained one would be four to six times below its detection floor.

The same measurement bounds what we may conclude. Whether faint retained content persists below

the floor is not answerable with this instrument at any distance beyond a few tokens, so this paper does not claim the state stops storing the image. It claims that what a probe reads at distance is not retention, and that it depends causally on the attention channel.

This resolves three things at once. It explains why the measured horizon exceeds the predicted one, and why the gap grows in smaller models: the probe has never been measuring retention. It explains why the fast heads appeared to retain most, since they track the residual stream most closely and the residual stream is where the re-supplied information arrives. And it supplies the mechanism behind the causal result, which the next section takes up.

8 Result 4: Causally, the Answer Follows the Attention Channel

The splice exchanges one memory channel between two runs that saw different images and then asks the host’s question. Pairs are admitted only if the host and the donor each answer their own question correctly, so the ceiling is 100% by construction. Of 111 attempted pairs, 11 were rejected on that criterion and none were rejected for mismatched cache length. Every recurrent swap was verified to have changed the state, 100 of 100.

condition	layers swapped	follows host	follows donor
no swap	0	100/100 [96, 100]	0/100 [0, 4]
recurrent, all 81 states	0	100/100 [96, 100]	0/100 [0, 4]
attention, 1 layer	1	100/100 [96, 100]	0/100 [0, 4]
attention, 3 layers spread	3	98/100 [93, 99]	0/100 [0, 4]
attention, 7 layers	7	4/100 [2, 10]	94/100 [88, 97]
attention, all 13	13	0/100 [0, 4]	98/100 [93, 99]
attention, earliest 3	3	94/100 [88, 97]	1/100 [0, 5]
attention, latest 3	3	100/100 [96, 100]	0/100 [0, 4]
both channels	13	0/100 [0, 4]	100/100 [96, 100]

Table 7: The splice at 100 pairs with a 100% ceiling. Brackets are 95% Wilson intervals in percent. Replacing the entire recurrent memory of all 81 layers changes nothing; replacing seven of the thirteen attention caches flips the answer to the other image.

Three readings. First, the recurrent null is now complete rather than approximate: exchanging all 81 recurrent states and convolution buffers leaves 100 of 100 answers following the host’s own image, with every swap verified to have taken effect. Second, the attention channel is causally sufficient: swapping seven of the thirteen caches moves 94 of 100 answers to the donor’s image. Third, and new, the dependence is graded and distributed. One attention layer does nothing, three spread across the stack move two answers, seven move almost all of them. No small subset suffices, so the picture is held across the attention layers rather than concentrated in a few. The earliest three layers have a small effect where the latest three have none, which is consistent with visual information entering early and being consumed downstream.

Read with Result 3, the recurrent null stops being a puzzle. Replacing the recurrent state cannot change the answer because the replacement is erased within a few tokens and rewritten from the attention channel, which the recurrent swap leaves untouched. The two results are the same fact seen from two sides.

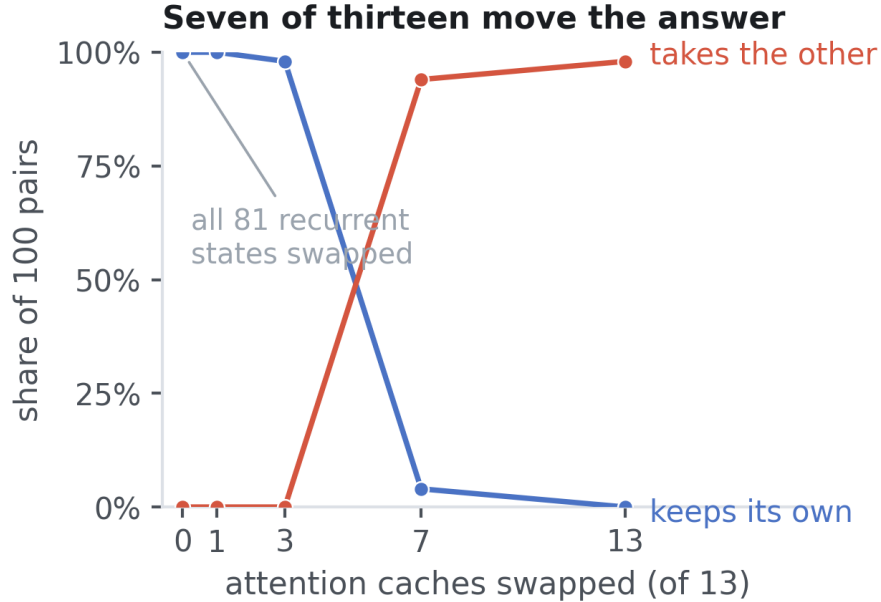


Figure 6: How many of the thirteen attention caches must be exchanged before the answer follows the other image, over 100 pairs in which both images were answered correctly beforehand. The leftmost point is the recurrent swap: all 81 states replaced, no attention cache touched, no effect.

9 Result 5: Scale, and Where the Prediction Fails

The instruments read every shape from the loaded model’s configuration rather than assuming Zamba2-VL-7B’s, which is necessary rather than tidy: the 2.7B and 1.2B checkpoints use a single B/C group where the 7B uses two, and the 1.2B carries a state dimension of 128 where the others carry 64. Table 8 reports the same protocol at all three scales.

model	layers	attn	probe $d=0$	probe $d=64$	measured $t_{1/2}$	predicted $t_{1/2}$
Zamba2-VL-7B	81	13	71.7%	21.7%	4.0	6.64
Zamba2-VL-2.7B	54	9	84.2%	31.7%	9.6	2.91
Zamba2-VL-1.2B	38	6	83.3%	34.2%	14.3	2.58

Table 8: The same protocol at three scales, 120 identity trials per distance. The pre-image control reads exactly 12.5% at every scale. Half-lives are in tokens: measured is the distance at which above-chance decodability halves, predicted is the median over heads of $\ln 2 / (|A|\bar{\Delta})$.

Two things follow, and the second is a negative result we report rather than bury.

The parameter-derived horizon is short at every scale. Median predicted half-lives are 6.64, 2.91 and 2.58 tokens, and the fraction of heads with a predicted half-life under 32 tokens is 85.0%, 97.9% and 95.6%. Whatever the recurrent state genuinely retains of its own past, it retains for single-digit token counts in all three models.

But the predicted horizon does not predict the measured one. Across the three models the two quantities move in opposite directions: prediction falls with scale while measurement rises. On the 7B alone the agreement looked strong, with measured decodability tracking predicted retention at $r = 0.973$ across eleven distances. That agreement does not replicate, and we therefore do not claim that the decay calculation explains the measured decay curve. The next section shows why it cannot, and what the

probe is actually reading at distance.

10 Result 6: Visual Key-Value Eviction Removes the Only Copy, Eventually

The redundancy premise behind visual token and key-value pruning can be tested directly. We evict a fraction of the visual key-value entries from all 13 attention layers immediately after the image is read, then let the filler and the question stream past the pruned cache, and measure the model’s own accuracy. Retention p is the fraction of visual positions kept, sampled uniformly, and the question is asked d tokens after the image.

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
100%	97.9%	95.8%	100.0%	97.9%
25%	91.7%	95.8%	95.8%	91.7%
5%	87.5%	77.1%	75.0%	64.6%
0%	64.6%	37.5%	31.2%	22.9%

Table 9: Behavioural accuracy under visual key-value eviction, against distance between the image and the question. $n = 48$ per cell, chance 12.5%, no failures. The unpruned row is flat, so distance alone costs nothing.

Read as cost relative to the unpruned baseline at the same distance, the structure is a clean interaction:

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
25%	6.2	0.0	4.2	6.2
5%	10.4	18.8	25.0	33.3
0%	33.3	58.3	68.8	75.0

Table 10: Accuracy points lost to pruning, relative to the unpruned baseline at the same distance.

At a quarter of the entries retained there is no distance effect at all: the cost is 6.2, 0.0, 4.2 and 6.2 points, which is flat within noise. The redundancy premise holds on this architecture at that level, and we found no evidence against it.

Below that level distance begins to matter, and it matters a great deal. At a twentieth retained the cost triples across the range, from 10.4 points beside the image to 33.3 points at 512 tokens. With every visual entry removed it runs from 33.3 to 75.0. The unpruned row is flat across the same distances, so this is not a difficulty that grows with prompt length; it is a difficulty that grows with prompt length *only once the visual entries are gone*.

There is a retention threshold, somewhere between a quarter and a twentieth on this model, below which the cost of pruning stops being a fixed overhead and starts growing with how far the question sits from the image. A method validated at short range will understate its cost at long range by a factor of three at 5% retention.

The zero-retention row is also the cleanest available measure of what the recurrent channel contributes on its own. With no visual entries anywhere in the attention caches, accuracy falls from 64.6% beside the image to 22.9% at 512 tokens. That decline is the recurrent channel’s contribution decaying, measured behaviourally and with an instrument entirely independent of the probe, and it agrees with the picture the probe gives.

There is a discrepancy worth stating plainly rather than smoothing over. At zero retention the model still answers at 37.5% at thirty-two tokens, well above the 12.5% chance rate, while our linear probe

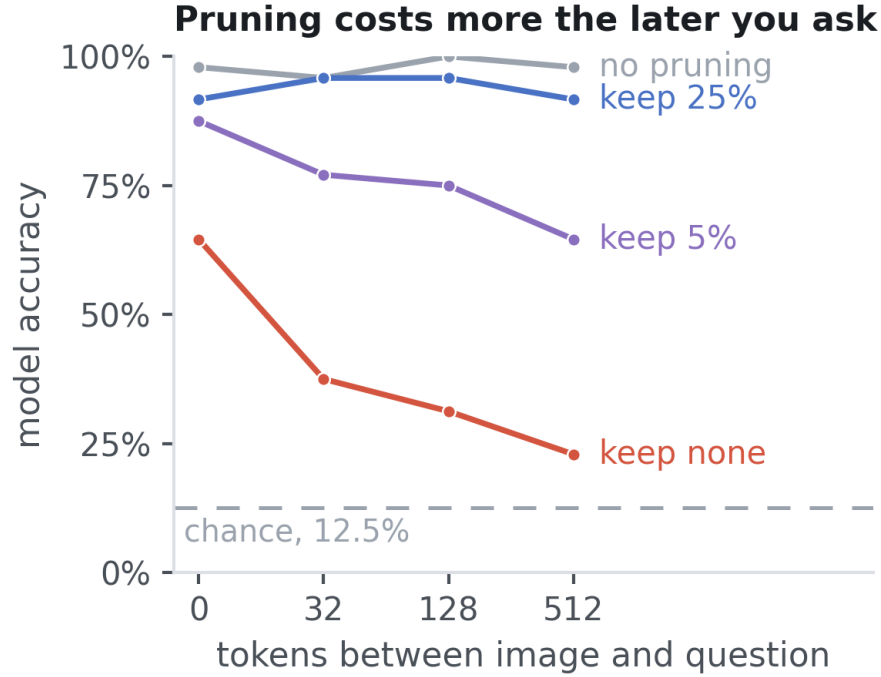


Figure 7: Model accuracy under visual key-value eviction, against how far the question sits from the image, $n = 48$ per point. The unpruned line is flat, so distance alone costs nothing. Keeping a quarter of the entries is also flat. Below that the lines fan out: the cost of pruning grows with distance only once the recurrent channel has stopped relaying the picture.

reads the recurrent state at close to chance at that distance. The model is using something in the recurrent channel that the probe does not recover. The probe is a lower bound on what the state carries, not a measure of it, and the sentence “the state is empty” is not one this evidence supports.

11 Limitations

We separate three kinds of limitation: alternative explanations our evidence does not exclude, limits on what the instruments can see, and limits on scope. The first kind is the one that should decide how much weight a reader puts on the central claim, so it comes first.

11.1 Alternative explanations we have not excluded

Eviction changes more than the image’s presence, though the matched control now argues against it. Removing the visual key-value entries shortens every attention layer’s cache by roughly 400 positions and changes the softmax normalisation for every token that follows. Section 7.1 reports the matched control, evicting an equal number of *non-visual* positions from the same caches, and finds the two interventions move the readout in opposite directions, which a pure perturbation cannot produce. What remains unexcluded is narrower: the pre-image text we remove is not interchangeable with the image in what it contributes to the residual stream, so the comparison bounds the perturbation account rather than eliminating every version of it.

The relay result could be a sensitivity asymmetry, now bounded but not eliminated. Our own data show the probe is a lower bound (below), and a lower bound need not be a uniform one: freshly re-

encoded content plausibly sits at higher amplitude than content written many tokens earlier and decayed. If the probe recovers high-amplitude content easily and low-amplitude content poorly, removing the re-encoded copy would collapse the readout even in a state still genuinely retaining something. The amplitude calibration of Section 7.3 answers this by measuring the detection floor and showing that predicted retention at thirty-two tokens sits below it, so a visible reading there cannot be a retained one. That argument assumes decay followed by overwriting acts on the discriminative component roughly as our attenuation does. A model with no attention channel to relay from would settle it without that assumption, and we have not tested one.

The route is located but not traced. The layerwise eviction of Section 7.1 shows the relay enters early: zeroing the three earliest attention layers reproduces 15.8 of the 17.2 points lost when all thirteen are zeroed, while the three latest produce 0.5. What is not traced is the intermediate path, from attention output through the residual stream into a downstream mixer’s write.

11.2 What the instruments can and cannot see

The probe is a lower bound, and our own behavioural data prove it. With every visual key-value entry removed, the model answers at 37.5% at thirty-two tokens while the probe reads the recurrent state near chance at the same distance. Something usable survives there that a 512-dimensional random projection, read by a linear probe trained on 120 examples, does not recover. Every statement in this paper about what the state contains should be read as a statement about what is linearly decodable under that instrument. This paper does not establish that the state is empty of the image at any distance, and makes no such claim.

Probe accuracy depends on the feature-to-trial ratio. Count read at chance with 128 projection features and reached 19.4% at 512. In the other direction, the full 41,472-dimensional state probe reads 71.7% where a single layer’s 512 features read 92.5% on identical trials. Absolute accuracies are therefore not comparable across collections of different size or width, and we report them only within a collection. The decay curve in Table 2 is internally consistent because the same 120 trials appear at every distance.

One projection basis for most results. The eviction contrast was repeated across three independent Gaussian projection bases and is stable under them (fast stratum at $d=32$: intact $31.2 \pm 1.6\%$ against evicted $9.9 \pm 1.8\%$, mean and standard deviation over seeds). The decay-curve collections still use a single fixed basis, so the variance attributable to the choice of basis is unmeasured there.

The stratum dose-response is measured on ten strata, in one model. With three thirds the eviction sensitivity is ordered exactly by predicted decay rate (37.5, 18.8 and 7.8 points, $r = 0.991$), which on three points is suggestive rather than established. Repeating it on deciles of predicted half-life gives $r = 0.800$ at 32 tokens and $r = 0.647$ at distance zero over ten strata, with about 6 points of binomial noise per cell at $n = 64$. The relationship therefore survives finer stratification, but it is measured on one model and one task.

11.3 Scope

One model family, and a second family we tried and failed on. Results come from three Zamba2-VL checkpoints, which is one vendor, one architecture family and one training recipe. This is the principal limit on the paper’s generality and we do not argue around it.

We attempted to extend the relay result to a second family and report the attempt because its failure is informative. Granite 4.0 h-tiny [36] is a Mamba2 hybrid from a different lineage, with 36 recurrent layers to 4 attention layers against Zamba2-VL-7B’s 81 to 13. Its decay parameters read out cleanly

and give a median predicted half-life of 8.7 tokens, so the horizon calculation of Section 5 transfers to a second family. The relay contrast does not, for an instrumental reason: on a code-word recall task the model answers correctly on 100%, 96.9% and 100% of trials at 0, 32 and 64 tokens, and yet a probe on its recurrent state reads at chance at distance zero. Scaling to 256 trials did not change this (13.7% against a 12.5% chance rate, $p = 0.31$), and neither did probing single layers, where the best of 36 layers reached 19.9%.

We suspected the item was too small to write a detectable trace, a one-token code word against a 345-token image, and tested that by repeating the carrier sixteen times so the item occupied 192 tokens. It did not help: 12.0%, 17.2% and 10.9% across the three strata. Repetition lengthens an item without diversifying what it writes, so the state ends up holding little more than the final repeat, which is not the manipulation we intended. The honest position is that we do not know why the instrument fails on this model, only that it does, and that no relay claim can be made beyond the Zamba2-VL family on this evidence. A vision-language hybrid from a second family, where the item is a real image, remains the right test and we have not run it. The sharper version of that test is a vision-language model with no attention layers at all, such as Cobra [32] or VL-Mamba [33], where probing the recurrent state must be measuring retention because there is nothing else it could be measuring. Both are Mamba1 models and would need the read operator ported before any number from them could be trusted.

One task format, one dataset, one modality. Eight-way forced choice on COCO buys an exact chance rate and a clean pre-image control, and costs generality to free-form answering, to other visual properties, and to whether any of this holds for text held in the same states.

The horizon is measured on one filler distribution. $\bar{\Delta}$ is input dependent. We measured it on natural prose and found the median half-life stable across images, from 6.44 to 6.86 tokens, but a filler with very different gate statistics would give a different horizon, which the same calculation would predict.

12 Conclusion

The recurrent state of this hybrid vision-language model is not a memory of the image in any useful sense. It is a selective summary of the picture’s gist with a lifetime of about four tokens, and that lifetime is not an empirical curiosity but a quantity computable from the model’s own decay parameters before any measurement is taken. What the model actually answers from lives in the thin attention channel. The practical consequence is that the visual key-value entries a hybrid carries are redundant only for as long as the recurrent state is still holding the picture, which is not long.

References

- [1] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. *ICML*, 2020.
- [2] A. Gu, K. Goel, and C. Ré. Efficiently Modeling Long Sequences with Structured State Spaces. *ICLR*, 2022.
- [3] B. Peng, E. Alcaide, Q. Anthony, A. Albalak, S. Arcadinho, S. Biderman, et al. RWKV: Reinventing RNNs for the Transformer Era. *arXiv:2305.13048*, 2023.
- [4] A. Gu and T. Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv:2312.00752*, 2023.
- [5] T. Dao and A. Gu. Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality. *ICML*, 2024.
- [6] M. Beck, K. Pöppel, M. Spanring, A. Auer, O. Prudnikova, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. xLSTM: Extended Long Short-Term Memory. *arXiv:2405.04517*, 2024.

- [7] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, et al. Jamba: A Hybrid Transformer-Mamba Language Model. *arXiv:2403.19887*, 2024.
- [8] P. Glorioso, Q. Anthony, Y. Tokpanov, J. Whittington, J. Pilault, A. Ibrahim, and B. Millidge. Zamba: A Compact 7B SSM Hybrid Model. *arXiv:2405.16712*, 2024.
- [9] S. De, S. L. Smith, A. Fernando, A. Botev, G. Cristian-Muraru, A. Gu, et al. Griffin: Mixing Gated Linear Recurrences with Local Attention for Efficient Language Models. *arXiv:2402.19427*, 2024.
- [10] L. Ren, Y. Liu, Y. Lu, Y. Shen, C. Liang, and W. Chen. Samba: Simple Hybrid State Space Models for Efficient Unlimited Context Language Modeling. *ICLR*, 2025.
- [11] R. Waleffe, W. Byeon, D. Riach, B. Norick, V. Korthikanti, T. Dao, A. Gu, et al. An Empirical Study of Mamba-based Language Models. *arXiv:2406.07887*, 2024.
- [12] S. Jelassi, D. Brandfonbrener, S. M. Kakade, and E. Malach. Repeat After Me: Transformers are Better than State Space Models at Copying. *arXiv:2402.01032*, 2024.
- [13] W. Merrill, J. Petty, and A. Sabharwal. The Illusion of State in State-Space Models. *ICML*, 2024.
- [14] K. Wen, X. Dang, and K. Lyu. RNNs are not Transformers (Yet): The Key Bottleneck on In-context Retrieval. *arXiv:2402.18510*, 2024.
- [15] S. Arora, S. Eyuboglu, A. Timalsina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré. Zoology: Measuring and Improving Recall in Efficient Language Models. *arXiv:2312.04927*, 2023.
- [16] S. Arora, S. Eyuboglu, M. Zhang, A. Timalsina, S. Alberti, D. Zinsley, J. Zou, A. Rudra, and C. Ré. Simple Linear Attention Language Models Balance the Recall-Throughput Tradeoff. *arXiv:2402.18668*, 2024.
- [17] G. Alain and Y. Bengio. Understanding Intermediate Layers Using Linear Classifier Probes. *arXiv:1610.01644*, 2016.
- [18] J. Hewitt and P. Liang. Designing and Interpreting Probes with Control Tasks. *EMNLP*, 2019.
- [19] Y. Belinkov. Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics*, 2022.
- [20] J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, S. Sakenis, J. Huang, Y. Singer, and S. Shieber. Causal Mediation Analysis for Interpreting Neural NLP: The Case of Gender Bias. *arXiv:2004.12265*, 2020.
- [21] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and Editing Factual Associations in GPT. *NeurIPS*, 2022.
- [22] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small. *arXiv:2211.00593*, 2022.
- [23] F. Zhang and N. Nanda. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. *ICLR*, 2024.
- [24] L. Chen, H. Zhao, T. Liu, S. Bai, J. Lin, C. Zhou, and B. Chang. An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models. *ECCV*, 2024.
- [25] S. Yang, Y. Chen, Z. Tian, C. Wang, J. Li, B. Yu, and J. Jia. VisionZip: Longer is Better but Not Necessary in Vision Language Models. *arXiv:2412.04467*, 2024.
- [26] Y. Shang, M. Cai, B. Xu, Y. J. Lee, and Y. Yan. LLaVA-PruMerge: Adaptive Token Reduction for Efficient Large Multimodal Models. *ICCV*, 2025.
- [27] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen. H₂O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. *arXiv:2306.14048*, 2023.
- [28] Y. Li, Y. Huang, B. Yang, B. Venkitesh, A. Locatelli, H. Ye, T. Cai, P. Lewis, and D. Chen. SnapKV: LLM Knows What You are Looking for Before Generation. *arXiv:2404.14469*, 2024.

- [29] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient Streaming Language Models with Attention Sinks. *ICLR*, 2024.
- [30] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual Instruction Tuning. *NeurIPS*, 2023.
- [31] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, et al. Qwen2.5-VL Technical Report. *arXiv:2502.13923*, 2025.
- [32] H. Zhao, M. Zhang, W. Zhao, P. Ding, S. Huang, and D. Wang. Cobra: Extending Mamba to Multi-Modal Large Language Model for Efficient Inference. *AAAI*, 2025.
- [33] Y. Qiao, Z. Yu, L. Guo, S. Chen, Z. Zhao, M. Sun, Q. Wu, and J. Liu. VL-Mamba: Exploring State Space Models for Multimodal Learning. *arXiv:2403.13600*, 2024.
- [34] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. L. Zitnick, and P. Dollár. Microsoft COCO: Common Objects in Context. *ECCV*, 2014. *arXiv:1405.0312*.
- [35] Zephyra. Zamba2-VL-7B model card. <https://huggingface.co/Zyphra/Zamba2-VL-7B>, 2026. Apache-2.0.
- [36] IBM Granite Team. Granite 4.0 h-tiny model card. <https://huggingface.co/ibm-granite/granite-4.0-h-tiny>, 2025.